

Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Streamlit	Chosen for fast development of data-centric web apps using Python, providing a responsive interface for audio recording buttons and clean analytics dashboards.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (FastAPI / Streamlit Backend)	Enables fast processing, seamless communication with AI models, and aligns with the team's engineering background.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite / PostgreSQL	Serves as a reliable relational database to easily map, organize, and store user profile logs, text transcripts, and feedback scores.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Streamlit Community Cloud	Offers quick, cost-effective hosting and instant deployment directly from our repository.
5	Version Control & CI/CD	GitHub	Ensures safe version control, effortless team collaboration, and a smooth backup of all code changes

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenAI Whisper API & LLM API	Whisper converts spoken audio into accurate text transcripts, while the LLM API analyzes technical accuracy, detects missing concepts, and tracks filler words like "um" or "uh".